

Logistic regression assumptions and diagnostics

Motivating example: Dengue data

Data: Data on 5720 Vietnamese children, admitted to the hospital with possible dengue fever. Variables include:

- + *Sex*: patient's sex (female or male)
- + *Age*: patient's age (in years)
- + *WBC*: white blood cell count
- + *PLT*: platelet count
- + other diagnostic variables...
- + *Dengue*: whether the patient has dengue (0 = no, 1 = yes)

Previously: Logistic regression model

Y_i = dengue status (0 = negative, 1 = positive)

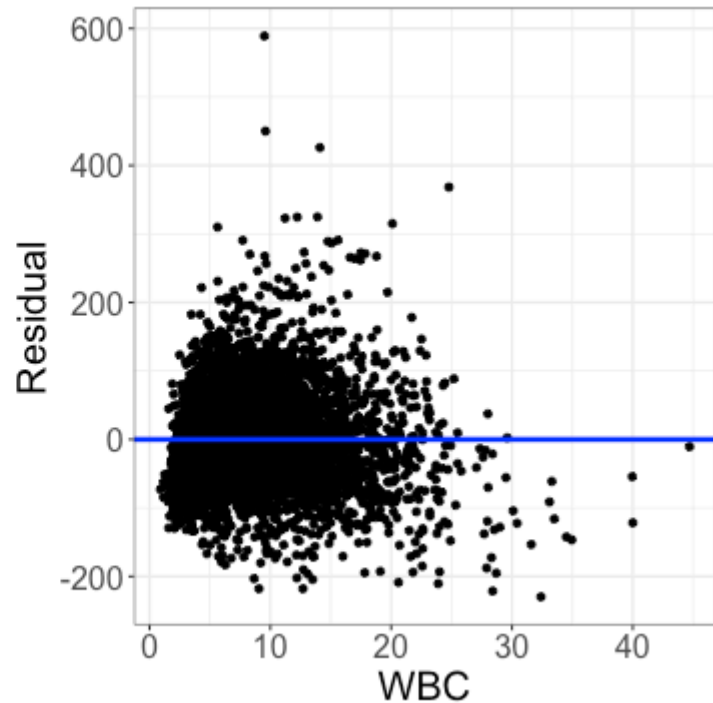
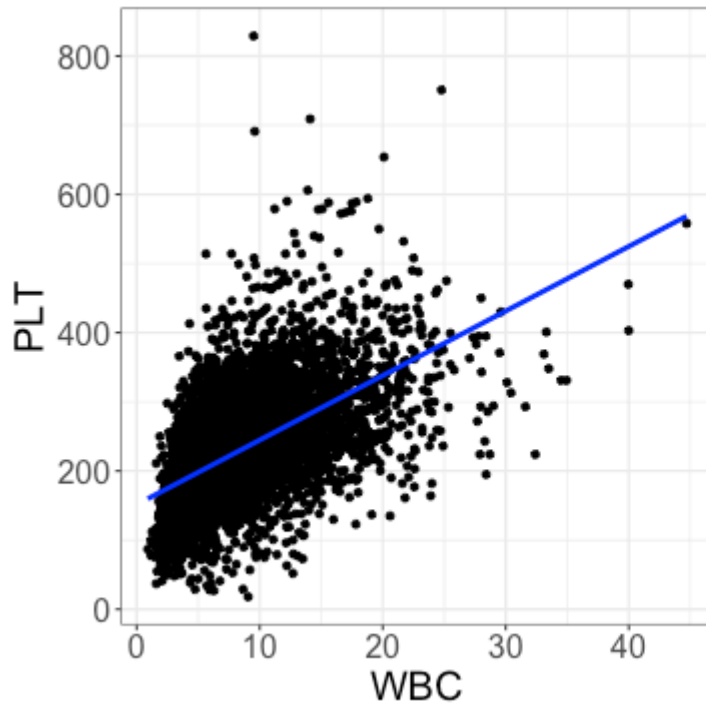
$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 WBC_i$$

What assumptions does this logistic regression model make?
How should we assess these assumptions? Discuss with your neighbor for 2--3 minutes, then we will discuss as a group.

Residual plots for linear regression

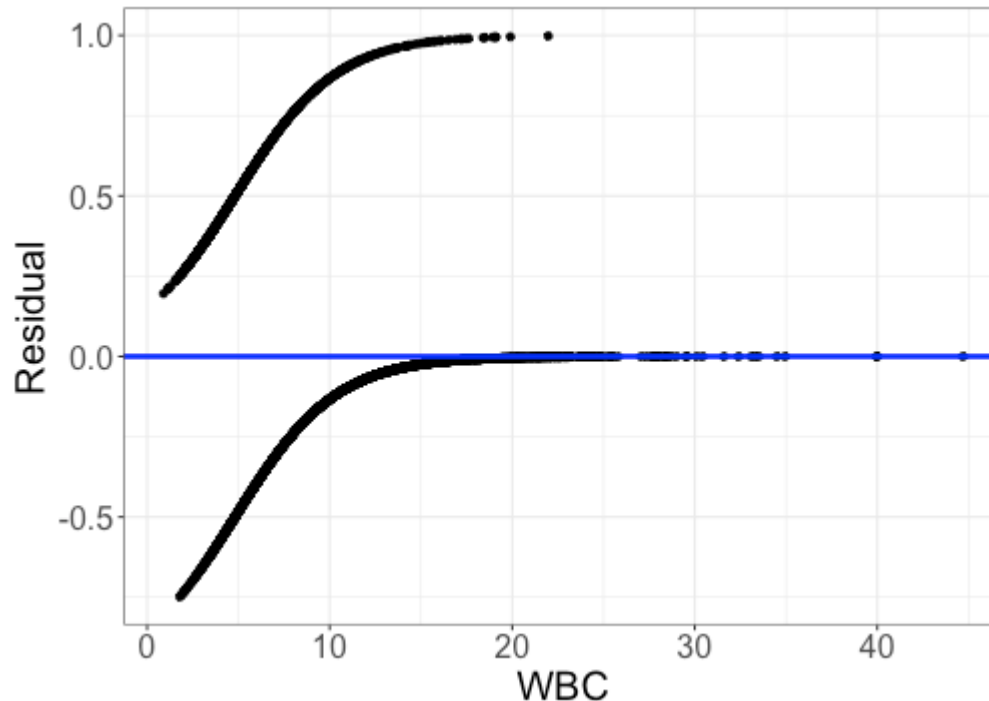
$$PLT_i = \beta_0 + \beta_1 WBC_i + \varepsilon_i$$



Don't use raw residuals for logistic regression

Fitted model: $\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = 1.737 - 0.361 \text{ WBC}_i$

Raw residuals $Y_i - \hat{\pi}_i$:



Assessing shape with empirical logit plots

Example: Putting data. Interested in the relationship between the length of a putt, and whether it was made:

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Length}_i$$

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134

Empirical logits

Step 1: estimate the probability of success for each length of putt

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134
Probability of success $\hat{\pi}$	0.832	0.739	0.565	0.488	0.328

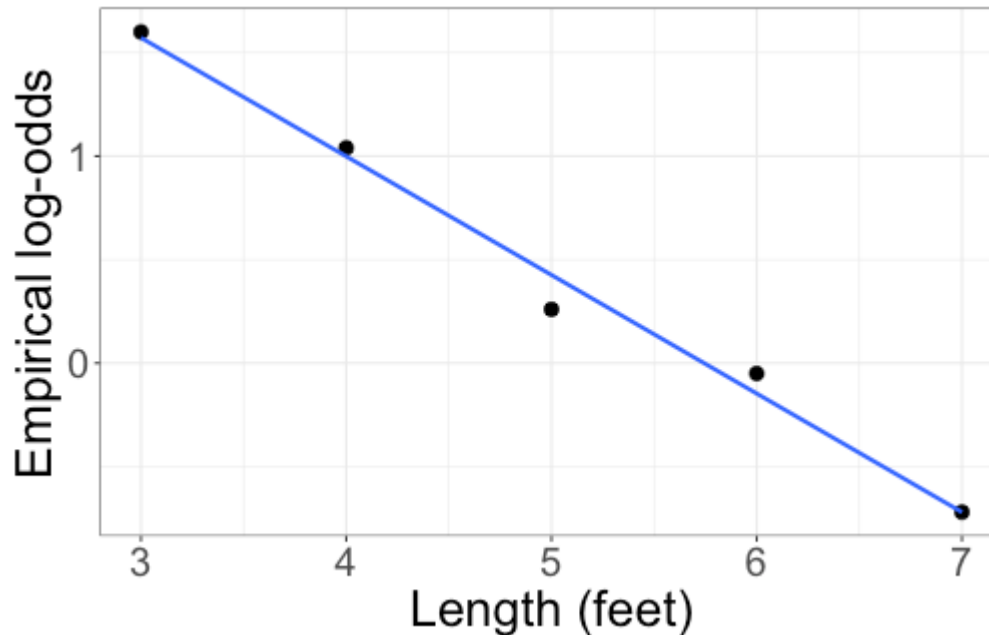
Empirical logits

Step 2: convert empirical probabilities to empirical log odds

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134
Probability of success $\hat{\pi}$	0.832	0.739	0.565	0.488	0.328
Odds $\frac{\hat{\pi}}{1 - \hat{\pi}}$	4.941	2.839	1.298	0.953	0.489
Log-odds $\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right)$	1.60	1.04	0.26	-0.05	-0.72

Empirical logits

Step 3: plot empirical log-odds against predictor, and add a least-squares line



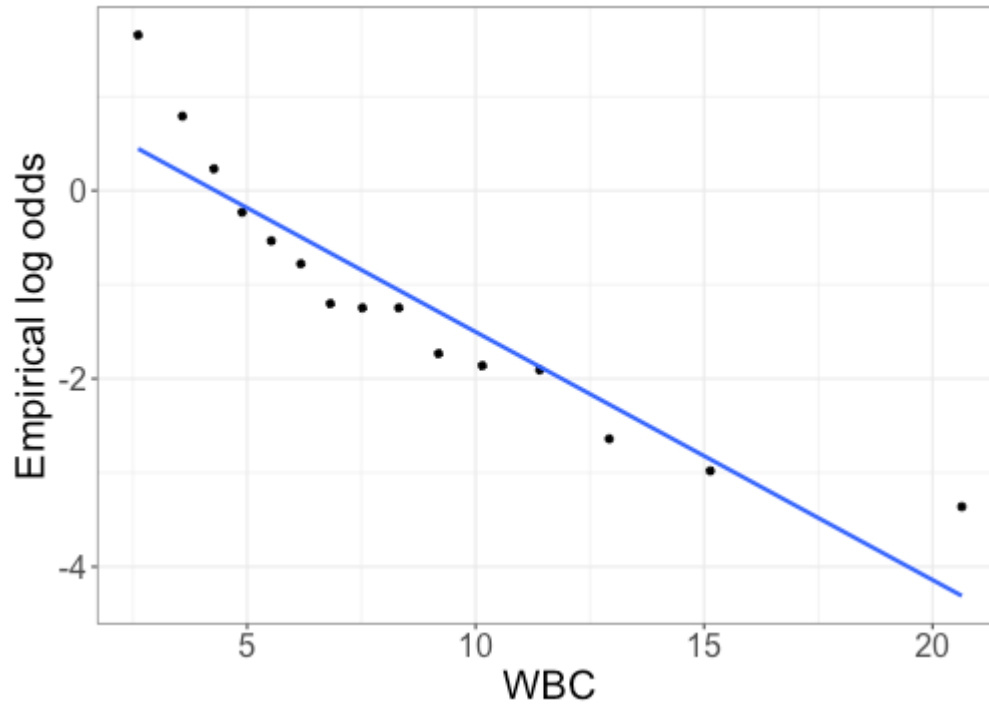
Does it seem reasonable that the log-odds are a linear function of length?

Back to the dengue data...

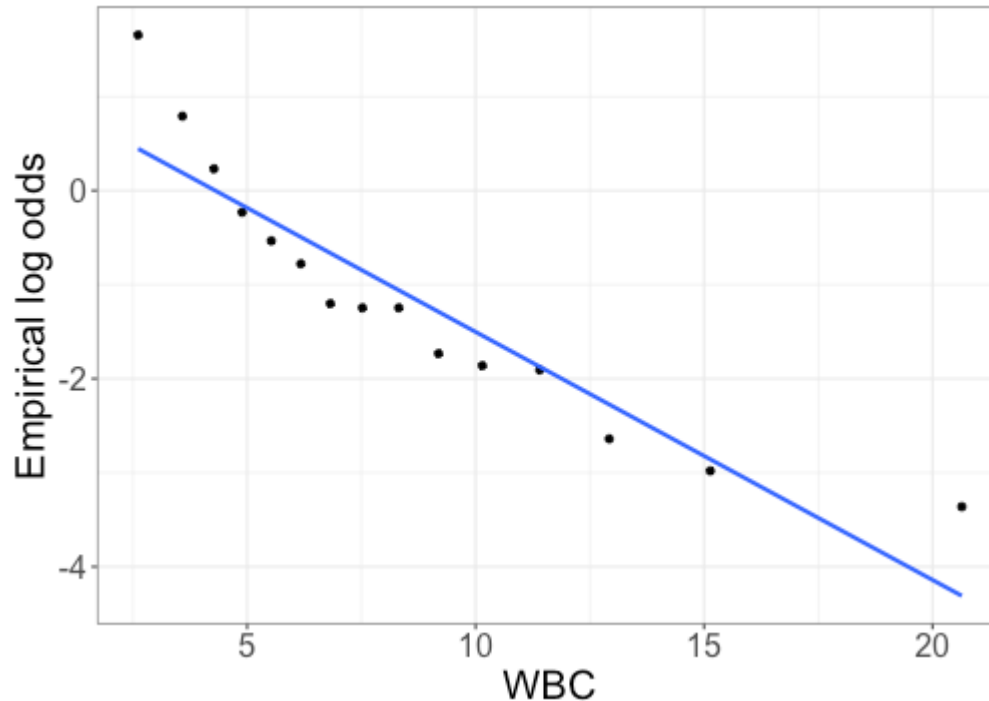
WBC	0.90	1.15	1.23	1.25	1.54	1.58	...
Dengue = 0	0	0	0	0	0	0	...
Dengue = 1	1	2	1	1	3	1	...

What problem do I run into?

Binned empirical logit plots



Binned empirical logit plots



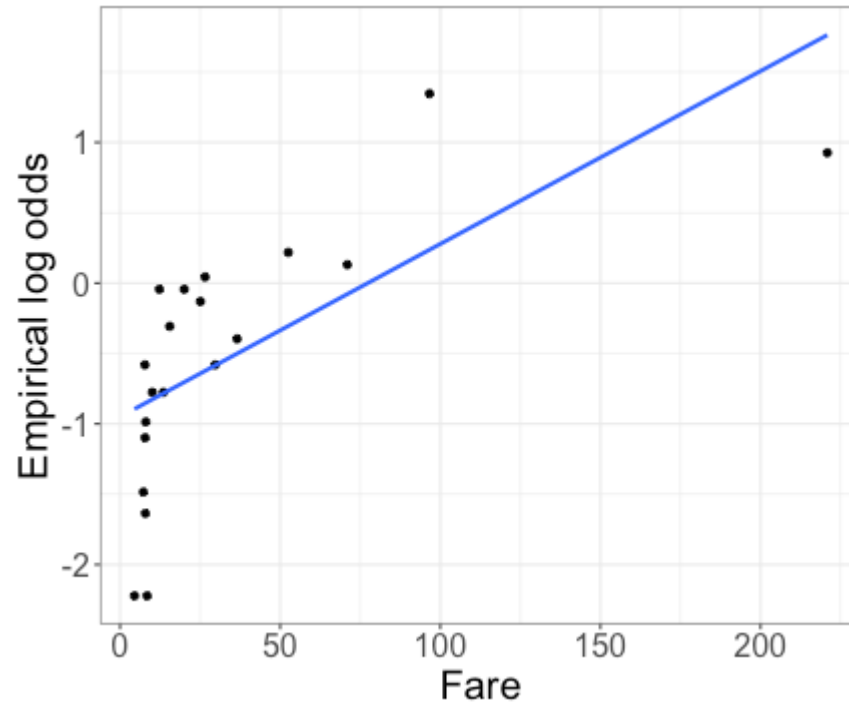
Does it seem reasonable that the log-odds are a linear function of WBC?

Class activity, Part I

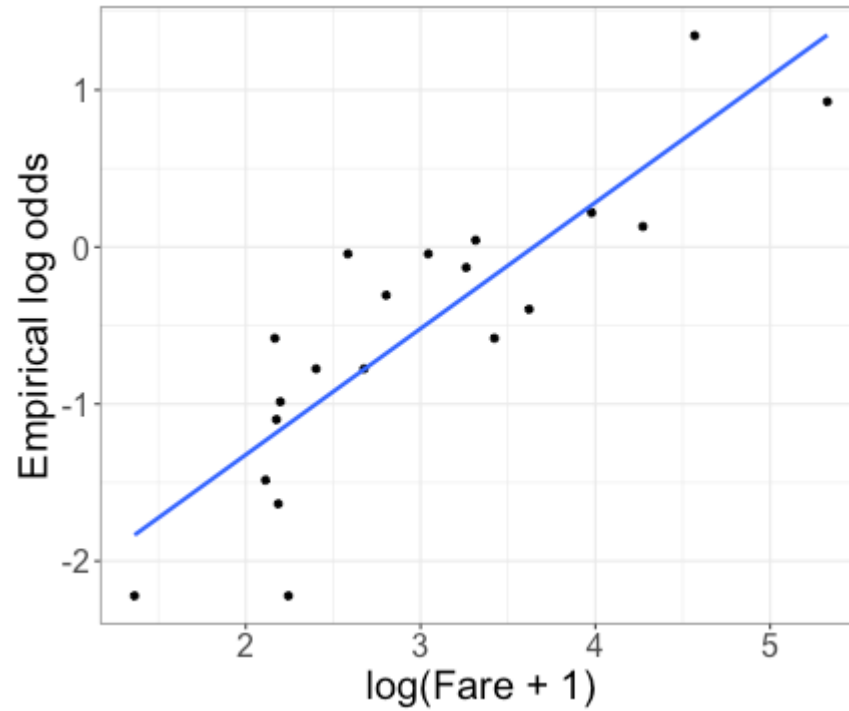
https://sta214-f25.github.io/class_activities/ca_07.html

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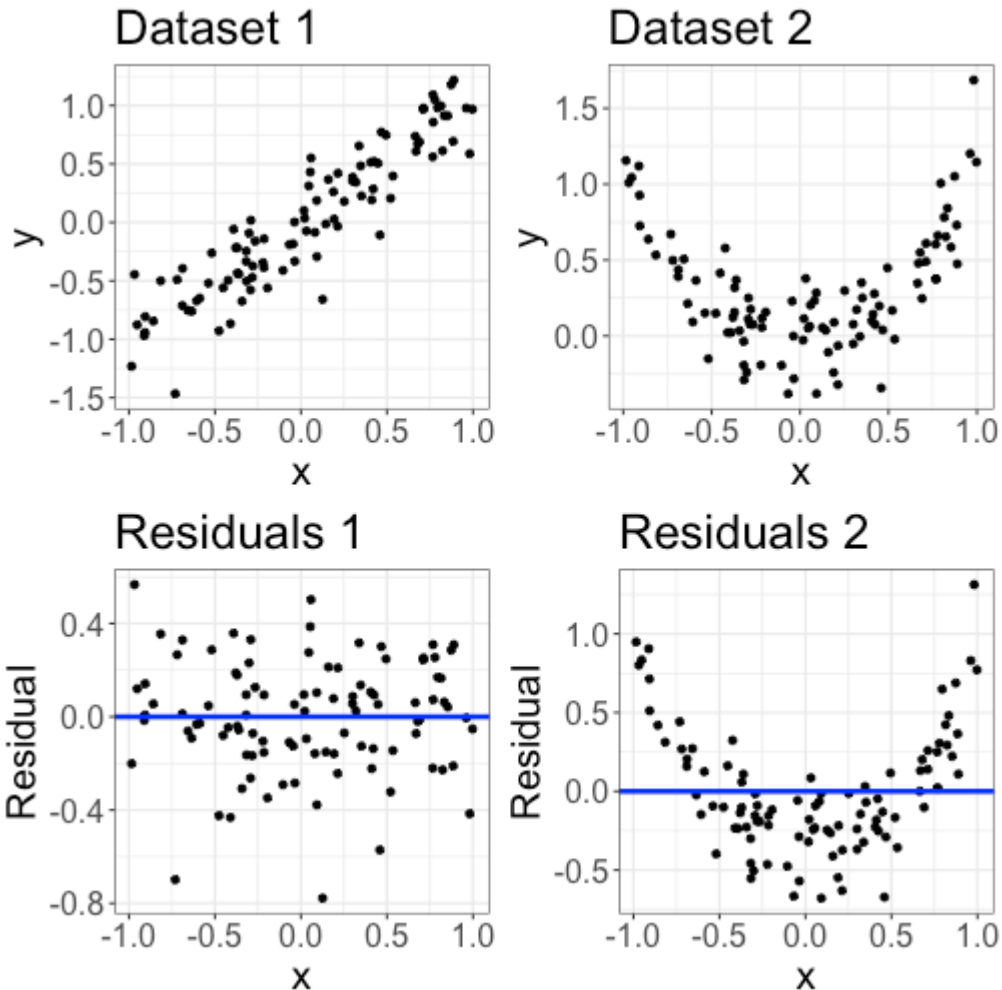
Class activity



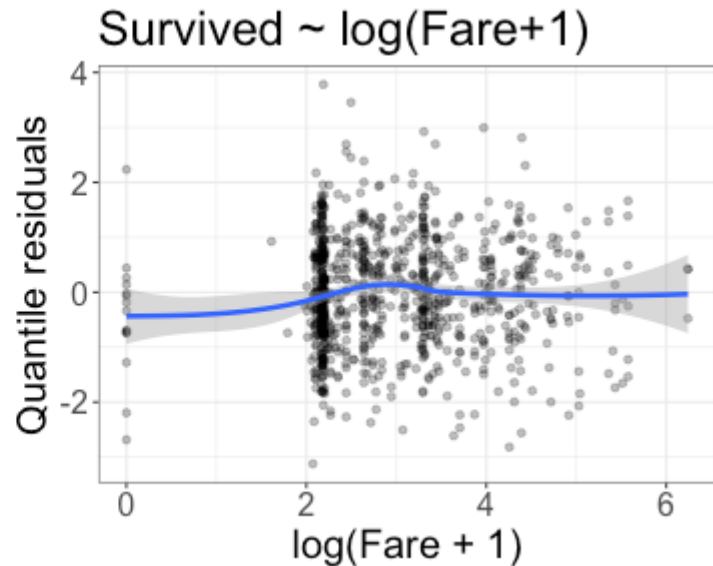
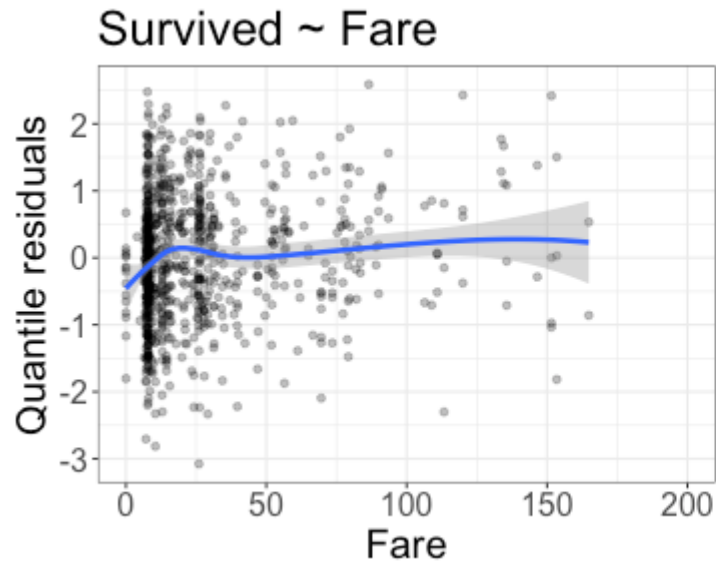
Class activity



Why residuals in linear regression are nice



Quantile residuals for logistic regression



Class activity, Part II

https://sta214-f25.github.io/class_activities/ca_07.html

(Don't need to submit this one)